09.07.25
Lab: 6

Implement Gradient Descent & Backpropagation in deep neural network.

Aim:

To implement gradient descent and backpropagation algorithm in deep neural network.

Objective:

1. To understand the working of gradient optimize
2. To implement backpropagation for updating neural network weights.
3. To observe the effects of iteration (epochs) on loss reduction

pseudocode

1. initialize weights and bias randomly
2. for each epoch:
   a forward pass:
      - compute weighted sum
        ($Z = wXX + b$)
      - Apply activation function
        ($A = F(Z)$)

   b. compute loss
      L = difference b/w predicted & actual

   c. Backward pass:
      → compute gradients of loss w.r.t weights &
      → update weights. $w = w - N * dL/dw$
      → update biases - $b = b - N * dL/db$

3, Repeat until convergence

formula used

$$z = w \cdot x + b$$

$$h = 1/n \sum (y - \hat{y}) z$$

$$w = w - \eta \cdot \frac{dC}{w}$$

Observation:

1, initially, the model started with random weights leading to high loss and low accuracy.

2, with each epoch, gradient descent gradually reduced the loss, showing effect on stimulus weight update

3, Backpropagation efficiently adjusted weights layer by layer.

4, The choice of learning rate and number of epoch strongly influenced convergence speed and final accuracy.

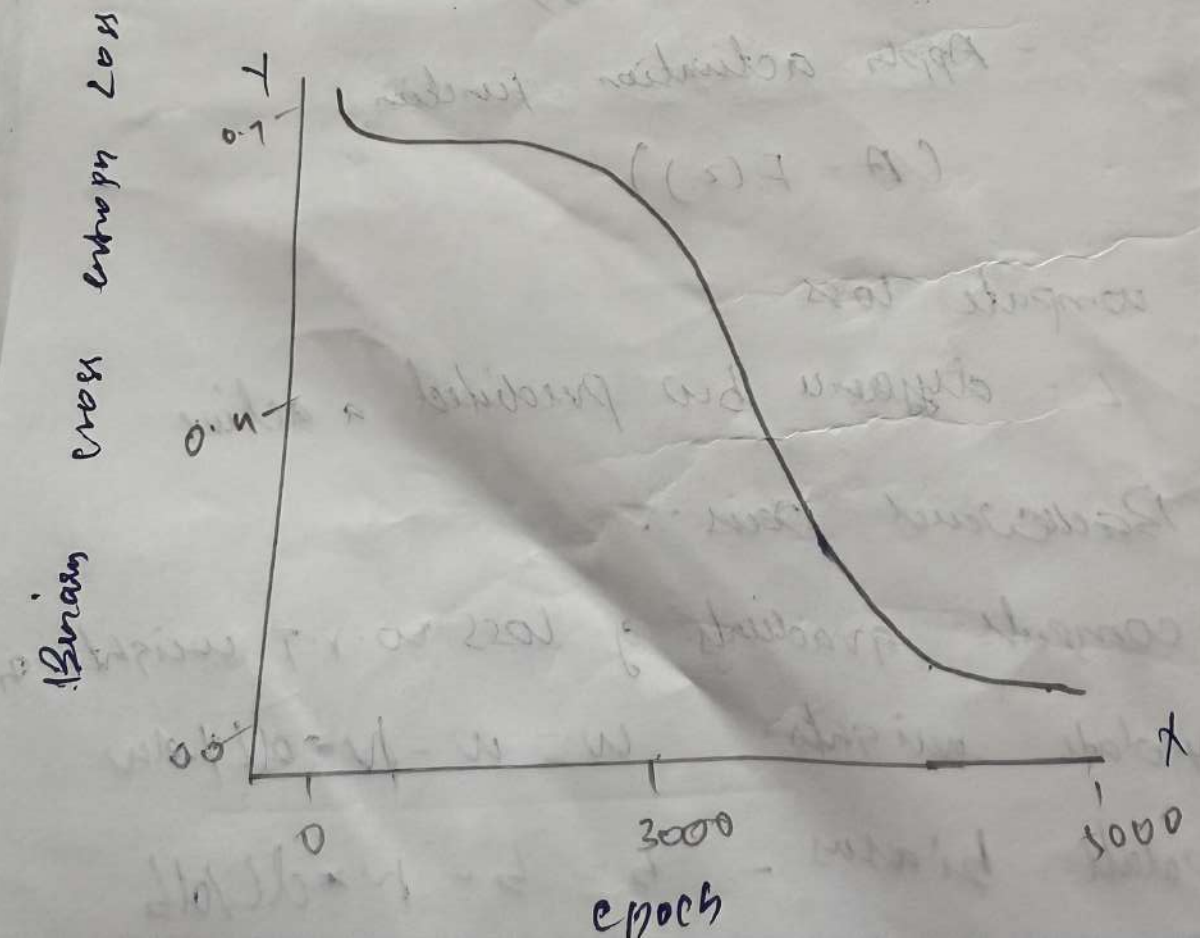
1, 9/9/25

Result:

implemented Gradient Descent & Backpropagation in DNN

epoch	Training loss
0	0.7198
500	0.6869
1000	0.6755
1500	0.6402
2000	0.5343
2500	0.2707
3000	0.0989
3500	0.0417
4000	0.0253
4500	0.0177

Loss over epochs



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Code | Notebook | Python 3 (ipykernel)

[1]:
```
import torch
import torch.nn.functional as F
import matplotlib.pyplot as plt
```

[2]:
```
X = torch.tensor([[0, 0],
                  [0, 1],
                  [1, 0],
                  [1, 1]], dtype=torch.float32)
y = torch.tensor([[0], [1], [1], [0]], dtype=torch.float32)
```

[3]:
```
torch.manual_seed(42)
input_size = 2
hidden1 = 4
hidden2 = 4
output_size = 1
```

[4]:
```
W1 = torch.randn(input_size, hidden1, requires_grad=True)
b1 = torch.zeros(hidden1, requires_grad=True)
```

[5]:
```
W2 = torch.randn(hidden1, hidden2, requires_grad=True)
b2 = torch.zeros(hidden2, requires_grad=True)
```

[6]:
```
W3 = torch.randn(hidden2, output_size, requires_grad=True)
b3 = torch.zeros(output_size, requires_grad=True)
```

[7]:
```
lr = 0.1
epochs = 5000
loss_history = []
```

[8]:
```
for epoch in range(epochs):
```

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Code | Notebook | Python 3 (ipykernel)

[8]:
```
for epoch in range(epochs):

    z1 = X @ W1 + b1
    a1 = torch.sigmoid(z1)

    z2 = a1 @ W2 + b2
    a2 = torch.sigmoid(z2)

    z3 = a2 @ W3 + b3
    y_pred = torch.sigmoid(z3)

    loss = F.binary_cross_entropy(y_pred, y)

    loss.backward()

    with torch.no_grad():
        W1 -= lr * W1.grad
        b1 -= lr * b1.grad

        W2 -= lr * W2.grad
        b2 -= lr * b2.grad

        W3 -= lr * W3.grad
        b3 -= lr * b3.grad

        # Zero the gradients
        W1.grad.zero_()
        b1.grad.zero_()
        W2.grad.zero_()
        b2.grad.zero_()
        W3.grad.zero_()
        b3.grad.zero_()
```

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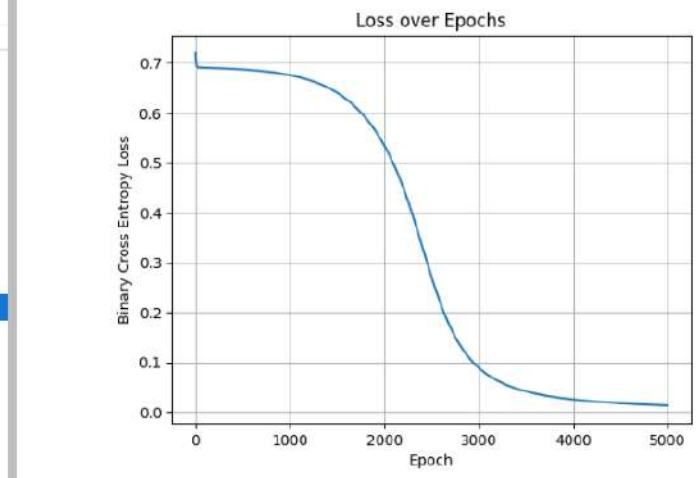
```
        W3.grad.zero_()
        b3.grad.zero_()

    loss_history.append(loss.item())


    if epoch % 500 == 0:
        print(f"Epoch {epoch} - Loss: {loss.item():.4f}")
```

```
Epoch 0 - Loss: 0.7198
Epoch 500 - Loss: 0.6869
Epoch 1000 - Loss: 0.6755
Epoch 1500 - Loss: 0.6402
Epoch 2000 - Loss: 0.5343
Epoch 2500 - Loss: 0.2707
Epoch 3000 - Loss: 0.0889
Epoch 3500 - Loss: 0.0417
Epoch 4000 - Loss: 0.0253
Epoch 4500 - Loss: 0.0177
```

[9]:
```
plt.plot(loss_history)
plt.title("Loss over Epochs")
plt.xlabel("Epoch")
plt.ylabel("Binary Cross Entropy Loss")
plt.grid(True)
plt.show()
```



[10]:
```
with torch.no_grad():
    z1 = X @ W1 + b1
    a1 = torch.sigmoid(z1)

    z2 = a1 @ W2 + b2
```

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Code | Notebook | Python 3 (ipykernel)

0 1000 2000 3000 4000 5000

Epoch

[10]:
```
with torch.no_grad():
    z1 = X @ W1 + b1
    a1 = torch.sigmoid(z1)

    z2 = a1 @ W2 + b2
    a2 = torch.sigmoid(z2)

    z3 = a2 @ W3 + b3
    y_pred = torch.sigmoid(z3)
    predicted = (y_pred > 0.5).float()

print("Predictions:\n", predicted)
print("Ground Truth:\n", y)
```

```
Predictions:
 tensor([[0.],
        [1.],
        [1.],
        [0.]])
Ground Truth:
 tensor([[0.],
        [1.],
        [1.],
        [0.]])
```

[ ]:

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